PROJECT SYNOPSIS REPORT

ON

PROPREP

SUBMITTED

TO

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FOR

INTEGRATED PROJECT(CS203)

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Problem Statement

Job seekers often face challenges in creating optimized resumes and preparing for interviews due to fragmented tools and a lack of personalized AI-driven insights. This project AI Mock Interview App offering seamless, real-time feedback for interview preparation. By leveraging AI, it streamlines the job application process, helping users enhance their career readiness and increase their chances of success in the hiring process.

Title of project:

ProPrep :

"Shaping the Future of Career Success with AI- Interview Support"

Objective & Key Learnings:

- **Develop a Seamless AI-Powered Mock Interview Tool:** Gained hands-on experience in creating a unified platform integrating both AI-driven mock interview functionalities, enhancing job seekers' career preparation.

- **Leverage AI for Personalization:** Learned to integrate advanced AI technologies (like Gemini AI) to provide tailored resume suggestions and adaptive interview feedback, enhancing the user experience and effectiveness of both tools.

- **Design an Intuitive User Interface:** Enhanced skills in UI/UX design by building a user friendly, interactive platform that fosters engagement and ease of use for diverse user needs.

- **Build a Full-Stack Application:** Developed expertise in full-stack development (React,

Next.js, Drizzle ORM) while building the frontend and backend for a scalable, secure platform.

Options available to execute the project:

Mock Interview Functionality (AI-Driven Feedback)

- **Gemini AI for Adaptive Feedback**: Leverage AI to provide personalized feedback based on interview responses.
- **WebSocket/Socket.io**: For real-time interaction between users and the system during mock interviews.

User Authentication & Management

- **Clerk for Authentication**: Secure user authentication and management using Clerk to handle sign-ups, logins, and profiles.

Backend Development (API & Data Management)

- **Next.js API Routes**: For handling backend logic, AI model integration, and interactions with the database.
- **Drizzle ORM**: For handling database interactions and ensuring smooth data management.

Data Storage and Management

- **Cloudinary or S3 for File Storage**: For uploading, storing, and retrieving user uploaded files (interview recordings).

Advantages:

- **AI-Driven Personalized Experience:** The integration of AI mock interview feedback provides personalized suggestions, making the platform more effective and tailored to each user's career needs.
- **Scalability:** The use of Next.js for backend and frontend development, along with Vercel and Heroku for deployment, ensures that the application can scale easily as user demand increases.
- **Real-Time Feedback:** The mock interview feature's real-time feedback, powered by AI, gives users immediate insights and helps them improve interview skills interactively.
- **Secure User Authentication:** Integration with Clerk ensures secure user management, authentication, and session handling, which is essential for safeguarding user data.
- **Cloud Integration for Storage:** With Cloudinary for file storage, users' resumes and interview data are securely managed and easily accessible.

Disadvantages:

- **Dependence on AI Accuracy:** AI-generated mock interview feedback may not always be perfect, as the models can misinterpret some user inputs or provide suboptimal advice, which could affect user satisfaction.

- **Data Privacy and Security Concerns:** While tools like Clerk ensure secure authentication, storing sensitive data (resumes, personal information) online always carries privacy risks, and securing this data against breaches is a constant challenge.

- **Dependency on Cloud Services:** The reliance on Cloudinary, MongoDB Atlas, and other thirdparty services can be a disadvantage if the cloud provider experiences downtime or changes their pricing model.

- **Resource Intensive**: Running AI-powered simulations and processing large datasets (for feedback) might require significant computational resources, increasing the cost of hosting and processing, especially for a growing user base.

- **Real-Time Interaction Challenges:** Implementing real-time interaction in the mock interview section with Socket.io or WebSocket may lead to performance issues if not properly optimized, especially if many users are interacting simultaneously.

References:

1. https://gemini.google.com

2. https://nextjs.org/

3. https://socket.io/

4. https://strapi.io/